

Project Requirement Document

PROJECT TITLE: "DevTrack / OmahTI Development Hub"
CASE: Human Development





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Team Structure

- Member:
 1. Christiano Jose Intoro
 2. Fahmi Abdillah Zain
 3. Kevin Antonio Wiyono Lauw
 4. Thomas Nadandra Aryawida
 5. Widad Muhammad Rafi



Project Overview

1. Executive Summary

This project serves to solve some of the issues present in Human Development in OmahTI, which are the lack of active contribution tracker, manual scheduling and attendance methods using outside applications or tools, and sifting feedback from members or even clients. The Objective of this project is to create a website which can solve these problems by creating a medium for people to track their personal progress, schedule and track attendance between members and seniors, and create a place to give feedback towards the organization. In turn we hope that using this product it may uplift spirit and encourage continuous growth in members and create a better structure for organization activities in the future using streamlined scheduling, attendance, feedback processing.

2. Design and Implementation Constraints

This project has some constraints that needs to be worked around, first and most importantly being the team's skill set limitation, as the members have limited knowledge and experience in UI/UX and frontend engineering. Another issue is the timeframe being relatively narrow when taking into account other activities and syncing member's schedules. Thus, the team will have to divide work efficiently to support weak aspects and keep the scope of the project in realistic size to keep up with deadlines.

3. Assumptions and Dependencies

Assumptions:

- Each member of the organization has unique emails to associate them in the system.
- Internet access is stable enough to access the website regularly
- Feedback given would be in a constructive and clear manner

Dependencies:

- Organization having a team to manage and respond to feedback given by members or outside sources.
- Availability of hosting channels to implement the website.
- Commitment from the team to act out each individual's part in creating the website



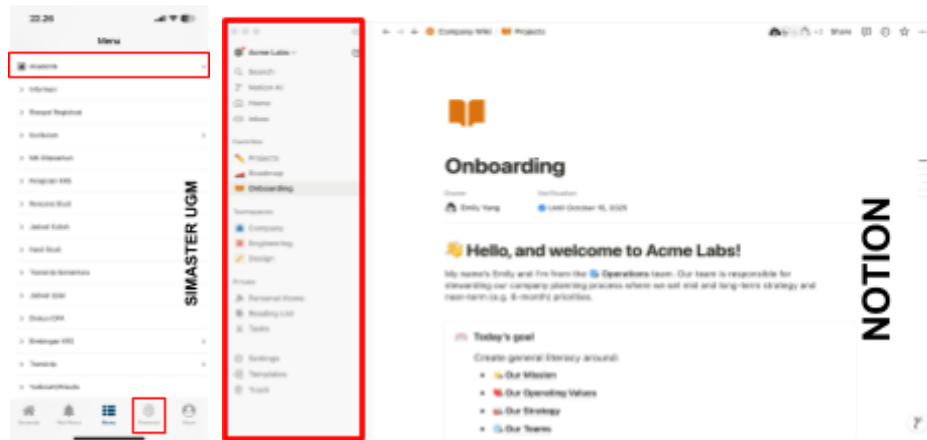
Project Requirements

4. Interfaces

4.1. User Interfaces

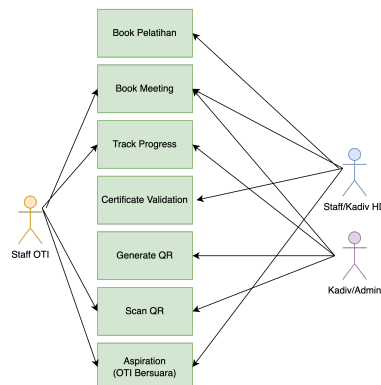
a. Design Style and Preference

The projects/websites that we work on take a lot of inspiration from existing brands, such as *notion*, *slack*, etc. Most of the features used in this website also take a lot of inspiration from the *simaster* website that we modified to suit the features that we want to adopt.



This website is also added with interactive components, such as robots that can follow the cursor, thus increasing the appeal of users in accessing this website. The design style of this website also embraces **human-centered based contemporary minimalism with professionalism**. The color palette used is also adjusted to OmahTI characteristics, namely orange and black, which add an elegant impression.

b. Use-case Decision

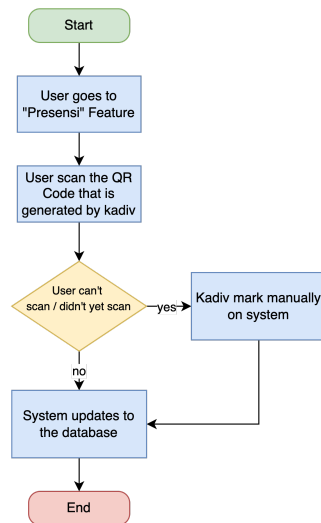




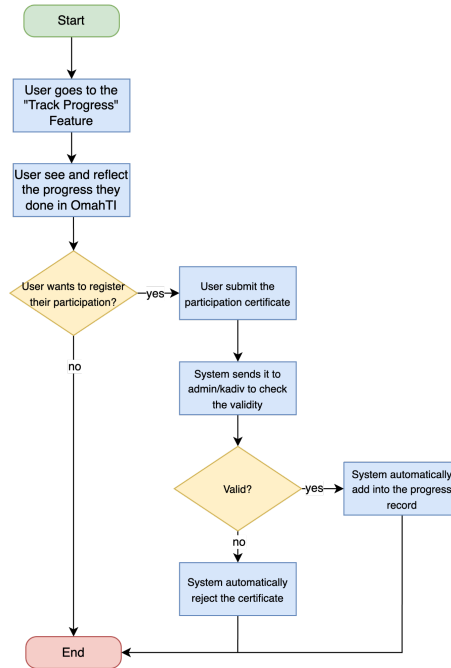
The use case diagram shows 3 actors who play the main roles in this project (Staff OTI, Staff/Kadiv HD, and Kadiv/Admin), where each with specific system access. Staff OTI can track their progress, scan QR, and submit aspirations through the "OTI Bersuara" feature. Staff/Kadiv HD can book meeting, track their progress, and can access the staffs and other kadivs aspiration through OTI Bersuara. Meanwhile, Kadiv/Admin has a lot of access to book meetings, track their progress, validating submitted staff's certificates, generate QR, and also scan QR. So that, the interfaces between a normal staff, kadiv, and HD Division are a bit different.

c. User Flow

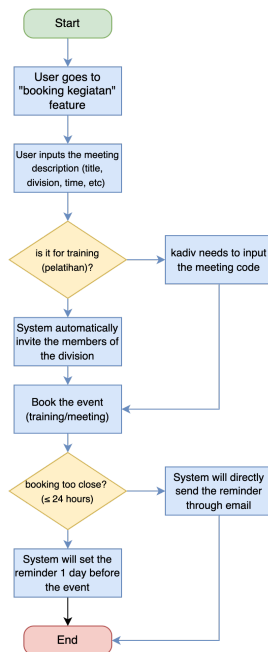
- **User Flow : Accessing *Presensi***



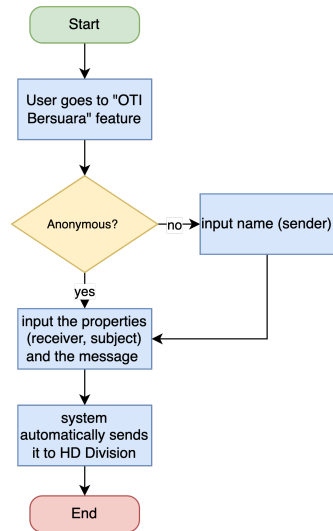
- **User Flow : Accessing *Track Progress***



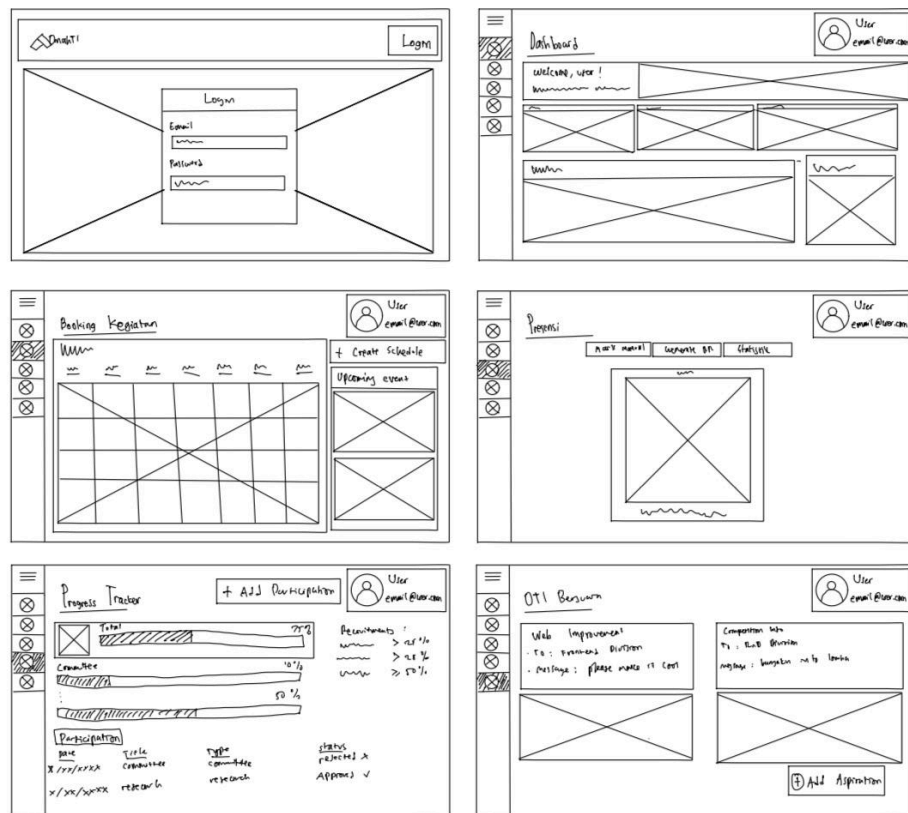
● User Flow : Accessing *Booking Kegiatan*



● User Flow : Accessing *OTI Bersuara*

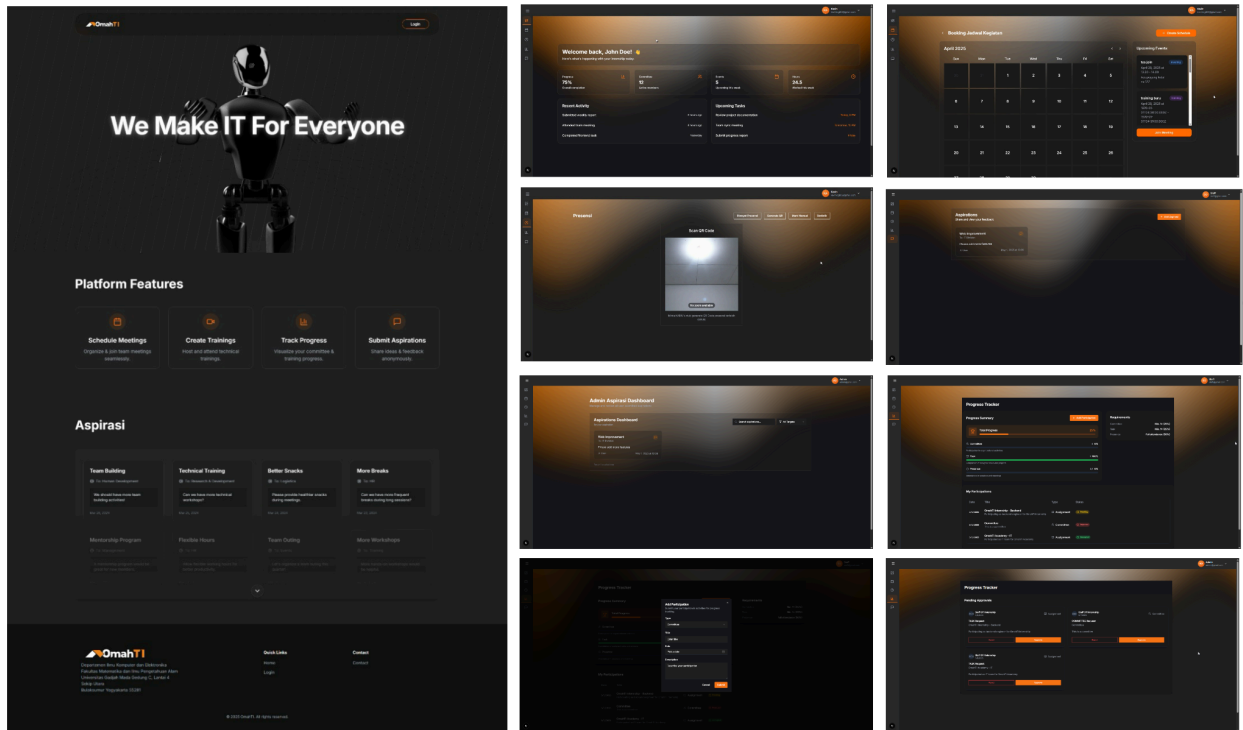


d. Wireframe





e. Mockup



4.2. Software Interfaces

- Frontend : **Next.js** and **TypeScript**
- Backend : **Express.js**
- Database : **PostgreSQL**
- Caching : **Redis**

Communication between each part of the system is done securely and efficiently. The frontend communicates with the backend using HTTPS, the backend accesses the PostgreSQL database through Prisma over a private connection, and it also talks to Redis through a direct internal connection. All backend-related components including the Express.js server, PostgreSQL database, and Redis are containerized using **Docker**. This means each of them runs in its own isolated environment, which makes development and testing more efficient. Docker ensures these services can work together without conflict, even across different machines or environments.

However, the system is currently only 50% deployed. The frontend has been successfully deployed and is accessible for users, but the backend has not yet been fully deployed online. This is because hosting the backend requires a VPS to support Docker and persistent backend processes, which involves high



costs. Due to budget limitations, the backend is currently run locally for development and testing purposes, while the frontend is live. Full deployment will be completed once a suitable hosting solution becomes available.

5. Functional Requirements

Functional requirements are essential to transform the domain problems faced by the Human Development division at OmahTI into concrete technical solutions. These requirements focus on the core functionality that must be provided through the website—such as managing staff, tracking attendance and participation, scheduling meetings, and handling aspirations. These requirements ensure that the final system is not only useful but also aligned with what the Human Development team needs to solve the problem at hand.

5.1. User Managements

- FR-1 : *System should be able to authenticate user*
- FR-2 : *System should be able to log in the user using with valid credentials*
- FR-3 : *System should support role-based access control (RBAC)*

6. Use-Case Requirements

- **User Authentication and Role-Based Access Control**

To maintain system security and ensure accurate data access, this website must include a secure authentication mechanism. Every user, whether regular staff, division head, or Human Development staff will log in using a verified email and password. This login system must implement role-based access control, which restricts or grants access to certain features based on the user's role.

7. Non-Functional Requirements

- **Caching and Optimization**

To improve performance, static content should be cached, and repeated queries (for example, retrieving a user's dashboard info) should be optimized using query caching or lazy loading mechanisms.

- **Error Handling and Logging**

The system must handle errors gracefully with clear messages to users and log detailed error information internally for diagnosis. Logging mechanisms should track system errors, failed logins, and potential misuse without violating user privacy.



Project Execution

8. Project Scope

This project aims to develop a centralized web platform to support Human Development in OmahTI by addressing several key challenges: the absence of an internal contribution tracker, the use of external and manual tools for scheduling and attendance, and a lack of structured feedback management. The platform will be designed to:

- **Member Progress Tracking** : Members can manually input their involvement in competitions, projects, and research. These contributions will be reflected in their personal progress dashboard and tracked for potential certification or recognition.
- **Presence Management** : The platform includes a QR-code-based attendance system that enables members to record their presence during training or events. Attendance data contributes to progress tracking and certification eligibility.
- **Schedule Booking and Notifications** : Members and administrators can create and manage bookings for training and meetings. The system includes smart reminders :
 - If the event is within 24 hours, a reminder is sent immediately.
 - If it's scheduled more than 24 hours ahead, reminders are sent 1 day before.
 - For training, invitations are automatically sent to all members within the relevant division.
- **OTI Bersuara** : A dedicated module enables members or external parties to submit opinions or suggestions to the organization. A centralized feedback system ensures transparency and encourages continuous improvement.

The platform will be used internally by OmahTI members, seniors, and administrators. It is scoped to a **web application**. The team will not integrate with third-party platforms or systems beyond email-based reminders. This scope corresponds to the commitments laid out in the internal agreement (MoU) between the internship team and OmahTI coordinators.



9. Project Deliverables

The development team is responsible for delivering a web-based platform tailored to internal organizational needs in OmahTI. The deliverables are broken down into functional modules, visual components, and documentation, as follows:

A web-based dashboard with :

1. Progress Tracking System

- Users can input and view their participation in competitions, projects, and research.
- Admins may view aggregated data for certification eligibility.

2. QR-Based Attendance System

- QR code generation and scanning for event attendance.
- Presence data will contribute to each member's overall progress.

3. Event Booking & Scheduling

- Booking system for Trainings and Meetings with input fields :
 - Title, Date, Time, Location
- Automated invitation and reminder system :
 - Training auto-invite all division members.
 - Meetings include a join code, with email reminders sent based on timing logic.

4. Feedback Module

- A public feedback form accessible via the homepage.
- Feedback entries are stored in a backend system and accessible by admin.



Supporting Deliverables with:

- **Pitch Deck**

A concise, visually appealing slide deck that summarizes the problem, solution, features, and impact of the project for presentation purposes.

10. Timeframe and Milestones

Phase	Duration	Dates
First Gathering and Case Division	1 day	17 March 2025
Mentoring	1 day	26 March 2025
UI/UX Design	2 days	April 20 - 22, 2025
Frontend Development	2 months	20 March - 1 May, 2025
Backend Development	2 months	20 March - 1 May, 2025
Final Deployment and Documentation	1 day	2 May 2025

11. Challenges and Open Questions

This part explains the problems we faced during the project and the things we are still unsure about that might affect the final result.

Challenges

- **Limited Project Duration:** The entire project is scheduled to be completed within one month, which requires efficient time and task management under a tight timeline.
- **Concurrent UTS Week:** The project overlaps with the midterm exam (UTS) period, limiting the team's available working hours and focus during critical development phases.



- **Inexperience in Web Development:** Several team members are still unfamiliar with frontend, backend, and UI/UX development, which could slow down progress and affect implementation quality.
- Etc.

Open Questions

Will there be a dedicated team within OmahTI to continue maintaining and improving the website after this internship ends?

How will user onboarding be handled to ensure that all members are familiar with the new system?

Will OmahTI officially integrate the attendance and progress tracking system for certification or evaluation purposes?

Should additional features (such as notification integration or admin moderation tools) be considered for future updates?



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